

LOKESH PATRA

+91 8249 83 6567 | lokesh0patra@gmail.com | linkedin.com/in/lokeshxlight | github.com/lightxlk
Cuttack, Odisha, India

EDUCATION

Sri Sri University <i>Baccalaureus Technologiae in CSE w/ Specialization in Data Science</i>	Cuttack, India Nov 2022 – Jul 2026
Stewart's Science Higher Secondary School <i>Intermediate Science</i>	Cuttack, India Aug 2020 – Apr 2022

PUBLICATIONS

- IEEE Xplore** | 6th Int. Conf. on Emerging Systems and Intelligent Computing (ESIC '26)
- Paper Title:** Roadkill Hotspots and Policy Advocacy (RHPA): A Multi-Layered AI and IoT Approach for Real-Time Wildlife Preservation.
 - Paper ID:** 1197 | **Achievement:** Awarded **1st Rank of Session**
 - Developed a 3-Dimensional framework combining Computer Vision (YOLOv5xu), Ensemble Deep Learning (DenseNet121 + EfficientNet-B3), and IoT to reduce wildlife-vehicle collisions.

EXPERIENCE

Indian Space Research Organization, Space Application Center <i>Research Scholar (BioSentinel Edge Project)</i>	Ahmedabad, India June 2025 – Present
- Engineered an Edge-AI vision system for low-power, real-time facial recognition achieving <500ms latency and <2W power consumption. - Optimized embedded inference using TensorFlow Lite and MobileFaceNet (INT8) on ESP32 hardware. - Developed a modular backend using FastAPI and Faiss vector search, implementing a hybrid cloud-edge architecture. - Utilized 100% GAN-based synthetic datasets for training to ensure zero biometric storage and strict privacy compliance.	
TSA Media Pvt. Ltd. <i>Associate SaaS Developer</i>	Cuttack, India February 2025 – Present
- Spearheaded the development of SaaS/Micro-SaaS platforms featuring interactive Web 3.0 interfaces. - Drove product growth from 10% to 45% monthly through the implementation of ML-driven insights and architectural enhancements. - Built predictive models and geospatial analytics pipelines, streamlining client operations via custom API backends.	
HackerRank <i>Software Engineering Intern</i>	Remote July 2024 – September 2024
- Contributed to production-level codebases adhering to strict SDLC standards and CI/CD workflows. - Collaborated on code reviews and version control using Git, improving module maintainability and debugging complex issues.	
Microsoft <i>Azure Intern</i>	Remote January 2024 – May 2024
- Designed large-scale data pipelines using the Azure ecosystem, performing ETL operations and data governance compliance. - Developed algorithms to generate actionable insights from high-volume datasets.	
TCS Academy <i>MasterCraft Intern</i>	Remote June 2024
Implemented data validation workflows and quality gates, enhancing system traceability and test data reliability.	

PROJECTS

- BioSentinel Edge** | *Python, TensorFlow Lite, ESP32, FastAPI, React*
- Built a real-time, ethical embedded facial recognition system running on microcontrollers with **<1MB RAM** usage.
 - Logged **120+ identities** and **300+ real-time events** during production testing using hybrid edge-cloud architecture.
 - Completed 7-phase development sprints covering ML, backend, frontend, and embedded optimization.
- Roadkill Hotspots & Policy Advocacy (RHPA)** | *YOLOv5xu, DenseNet121, IoT, MQTT*
- Designed a multi-layer AI + IoT system achieving **95% overall system accuracy** (93% detection, 97.5% classification).
 - Trained YOLOv5xu on **20,000+ annotated frames** and an ensemble of DenseNet121/EfficientNet-B3 to distinguish 'Alive' vs 'Dead' subjects.
 - Deployed ESP32-based actuation layer with ultrasonic sensors achieving **<0.5s alert response time**.
 - Pilot testing indicated a **36% reduction** in animal-vehicle collisions and achieved **92% hotspot overlap** with reported incidents.
- Predictive Geospatial Intelligence System** | *Cloud APIs, Geospatial Analytics*
- Built location-aware ML models for client applications using geospatial datasets for route scoring and event prediction.

TECHNICAL SKILLS

Languages: Python, SQL, LaTeX, ABAP, HTML/CSS, React

Machine Learning: TensorFlow, PyTorch, YOLOv5xu, OpenCV, Scikit-Learn, Pandas, NumPy, GANs

Backend & Cloud: SAP ABAP Cloud, FastAPI, Flask, Docker, REST APIs, Microsoft Azure, AWS

Hardware/IoT: ESP32, Embedded Inference (TinyML), Sensors (Ultrasonic/DHT), MQTT

Tools: Git, GitHub, CI/CD, Matplotlib, Seaborn, Jupyter

CERTIFICATIONS & AWARDS

SAP Certified Associate - Back-End Developer - ABAP Cloud

Issued: Nov 2025

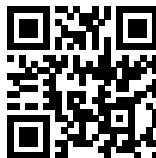
Validates ABAP RESTful Application Programming Model on SAP BTP and S/4HANA Cloud extensions.

1st Winner – IBM Innovation Center for Education '25

1st Runner-Up – IBM Innovation Center for Education '24

1st Runner-Up – OpenWeaver Sentiment Analysis Makeathon

Certified Software Engineering Intern – HackerRank



Thank You